



**MEFAMDEV-LIFE: A WEB-BASED MANAGEMENT AND SCHOLAR
LIFECYCLE MONITORING SYSTEM FOR FR. MARIE EUGENE
FAMILY DEVELOPMENT FOUNDATION**

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RECOMMENDATION SHEET

The capstone project entitled **MEFAMDEV-LIFE: A WEB-BASED MANAGEMENT AND SCHOLAR LIFECYCLE MONITORING SYSTEM FOR FR. MARIE EUGENE FAMILY DEVELOPMENT FOUNDATION**, prepared and submitted by Jerico Jimenez, Jaycee Marco, Reyjen Bong Sumaway, Angeline Toribio, and Ma. Karla Andrea Ubaldo in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval for oral examination.

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Internationally, there has been a rapid development in information and communication technologies and data analytics, which has revolutionized how organizations use data, however, this has not automatically or uniformly occurred equally across all sectors (Orero-Blat et al., 2024). According to Godefroid et al. (2023), there are many foundations and NGOs that have not taken advantage of this revolution in using digital data. They are still using paper-based or manual systems for record-keeping, reporting, and program delivery.

In the Philippines, community-based foundations have an important role to play in poverty and inequality reduction in areas because they operate at the grassroots level, directly engaging with vulnerable communities and tailoring interventions to local needs (Jumlail & Valerio, 2024). These foundations handle intricate information concerning their beneficiaries like family socio-economic profiles, academic performance data, values formation program attendance records, and program outcomes over time. According to Sarjito (2025), timely data access is not just a technical issue, it directly determines whether social assistance programs achieve their goals.

MEFAMDEV-Life is a proposed web-based management and scholar lifecycle monitoring system designed specifically for the Marie Eugene Family Development Foundation (MEFAMDEV), a community foundation registered with the DSWD and based in Angat, Bulacan. MEFAMDEV-Life is envisioned to manage the records of the

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foundation's beneficiary families and its recipients in elementary to Senior High School levels. It is envisioned to automate the monitoring of recipients' academic performance and spiritual formation attendance and serve as a data-driven tool for monitoring recipients' academic progression over time for the foundation's four staff members. By harnessing modern web technologies such as React.js for front-end and Laravel/PHP and MySQL for back-end and data management, MEFAMDEV-Life is envisioned to be accessible and usable by the foundation's staff in any internet-enabled devices, allowing its staff to manage recipients' and beneficiaries' records in the office and in the field.

1.2 Background of the Study

MEFAMDEV was founded by Notre Dame de Vie (NDV) institute or Ina ng Buhay, a secular institute that lives the Carmelite spirituality in 2006. This institute is a testimony of the desire of the members to be true to the teachings of their founder, Blessed Fr. Marie Eugene of the Child Jesus, OCD, on witnessing the life and love of God to men and women. The foundation is managed by four full-time staff members: the Director, who also serves as the primary Social Worker; an Education Social Worker; a Finance Officer/Bookkeeper; and a Program Coordinator, supported by volunteers.

MEFAMDEV is an agency with license number DSWD-FO III RL-0000024-2012, and it is accredited and certified by its license number DSWD-SB-A-0044-2019. The four core programs MEFAMDEV have are education, feeding, human development and values formation, and spiritual Christian formation. Out of the four programs they have, education is the heart of the program as well as the

family. MEFAMDEV relies on the core programs they have to help each family have enhanced value and life skills, personal growth, enriching prayer life, and Christian Faith.

MEFAMDEV has a total of 200-250 families it serves to this day, including the 103 total recipients, 75 are enrolled in the Basic Education levels (Elementary to SHS) and are the primary users of this system. The current manual system being used by the agency is that they are currently using Microsoft Excel and Microsoft Word for all their records. At the end of the year they manually tallied each attendance for formation sessions and is recorded on a paper. They currently have Computers and Laptops donated by some kind hearted people, but some of them are older models that struggle with Windows 11 updates. While the office has internet access, staff generally conduct their record-keeping offline, as the beneficiaries they serve in the field often have limited connectivity and access to digital devices. A typical day of data management for staff is monitoring grades every semester ends, tracking attendance by manually checking the signatures on a paper, and assisting over 75 recipients for requirements is tedious using only Excel and paper files.

1.3 Project Context

The development of MEFAMDEV-Life: A Web-Based Management and Scholar Lifecycle Monitoring System is a direct response to three specific and interrelated operational challenges identified within MEFAMDEV's current data management practices through interviews with the foundation's Director and staff, direct observation of daily workflows, and a review of existing documentation.

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The records of MEFAMDEV's beneficiary families and its Basic Education recipients from Elementary, Junior High, and Senior High levels are currently maintained in physical folders stored in confidential cabinets and in Excel files saved on office-bound computers. When staff members conduct off-site community visits in areas like Enkanto or San Roque, they are unable to retrieve or update beneficiary information in the field. This inaccessibility creates gaps in service delivery and increases the risk of data loss or inconsistencies when field observations must wait to be manually reconciled with office files.

Staff members must manually review the grade records of the 75 basic education recipients to identify those at risk of losing scholarship eligibility, a process that is currently tedious and error-prone. Monitoring spiritual and values formation attendance similarly requires manual cross-referencing of paper logs, which are often only tallied at the end of the year. This reactive approach means that by the time a struggling student is identified, the opportunity for timely intervention (addressing underlying issues like food insecurity) may have already passed, leading to wasted resources and budget.

While the foundation has a clear goal of integral development, it currently lacks a structured mechanism for tracking the academic trajectory of its recipients as they progress from Elementary through Senior High School. Because individual records are scattered across various Excel sheets and documents, there is no consolidated view of a student's performance over several years. This absence of longitudinal data prevents the foundation from identifying long-term patterns of improvement, recognizing

high-performing recipients, or generating evidence-based reports on the cumulative educational impact of the program before students reach the college level.

MEFAMDEV-Life addresses these three challenges through the following proposed solutions: a centralized, web-based management portal that digitizes student profiles and family intake forms for the Elementary to Senior High levels; an automated monitoring module that evaluates student grades and formation attendance against defined thresholds to generate immediate alerts for at-risk students; and a lifecycle tracking module that provides a consolidated longitudinal view of each student's academic and spiritual development throughout their enrollment in the basic education program.

1.4 Statement of the Problem

The study aims to answer the following questions:

1. How can the foundation eliminate data silos by transitioning the records of its beneficiary families and Basic Education recipients into a centralized web portal that ensures real-time accessibility for authorized staff members from any internet-enabled device?
2. How can the system automate the monitoring of academic grades and spiritual formation attendance of Basic Education recipients, while providing a consolidated year-over-year view of each recipient's academic progress, to replace manual, delayed, and error-prone audits with timely digital interventions?

3. How can the system enforce secure, role-based access controls to ensure that sensitive family and recipient data is protected from unauthorized access and that each user can only access information appropriate to their designated role?
4. How can the system enable authorized staff to generate accurate and timely reports on recipient academic performance and program outcomes to support internal program assessment and stakeholder reporting?
5. How can the system provide recipients with a secure, self-service portal to view their own academic records, spiritual formation attendance, and alert status, promoting transparency and personal accountability in their scholarship standing?

1.5 Objectives of the Study

This study aims to achieve the following specific objectives in direct response to the identified problems:

1. To develop a centralized web portal that digitizes and consolidates the socio-economic profiles of beneficiary families and the academic records of Basic Education recipients from elementary to Senior High School level, ensuring that authorized staff members can securely access and manage records in real time from any internet-enabled device.
2. To implement an automated monitoring module that tracks the quarterly academic grades and monthly spiritual formation attendance of all active recipients, featuring configurable digital triggers that automatically flag recipients who fall below defined performance or attendance thresholds, and a year-over-year academic growth tracking module that consolidates each recipient's grade records

across multiple school years into a unified longitudinal profile for timely and data-driven intervention.

3. To enforce a role-based access control mechanism that restricts system access according to each user's designated role, ensuring that sensitive family and recipient data is protected from unauthorized viewing, modification, or deletion at all times.
4. To develop a report generation module that enables authorized staff to produce summaries of recipient academic performance and program outcomes, providing data-driven insights for internal program assessment and external stakeholder reporting.
5. To provide recipients with a secure, self-service portal through which they can independently view their own academic records, spiritual formation attendance history, and current alert status, promoting transparency and personal accountability in their scholarship standing.

1.6 Significance of the Study

The development of the proposed system, MEFAMDEV-Life, for Fr. Marie Eugene Family Development Inc. (MEFAMDEV) aims to modernize the administration of student and beneficiary records. By transitioning from manual methods to a centralized digital platform, the study will provide significant advantages to the following:

MEFAMDEV Staff. The proposed system will streamline administrative workflows by replacing fragmented paper files and MS Excel spreadsheets with an organized digital repository. This allows staff to track academic histories and educational requirements in

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real-time, significantly reducing manual labor. Furthermore, the system will enable more efficient monitoring of student progress, ensuring that all records are up to date and easily accessible for reporting.

Recipients. The recipients will benefit from enhanced monitoring of their academic performance and extracurricular participation. Through the MIS, the organization can implement timely interventions; if a student's grades or attendance begin to decline, the system alerts staff to provide necessary support, such as tutorials or follow-up consultations. This proactive approach ensures that recipients remain focused and complete their education successfully.

Beneficiary Families. Beneficiary families will experience improved service delivery due to more accurate socio-economic profiling and data management. An organized system allows MEFAMDEV to better understand the specific needs of each household, ensuring that community programs and educational support are distributed more effectively and equitably.

Future IT Researchers. This study will serve as a valuable reference for future IT students and researchers, particularly those focusing on Information Systems for non-governmental organizations (NGOs) in the Philippines. It provides technical insights into designing secure and user-friendly systems tailored for organizations transitioning from offline, low-technology processes to automated environments.

1.7 Scope and Limitations

The study is focused on the design and development of MEFAMDEV-LIFE: A Web-Based Management and Scholar Lifecycle Monitoring System for the Fr. Marie

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Eugene Family Development Foundation. The system covers the essential functions of scholarship administration, including the digitization of socio-economic profiles, academic grade monitoring, and attendance tracking for spiritual formation and livelihood sessions. It is designed to serve three primary stakeholders: the foundation's administrators, authorized staff personnel, and the student beneficiaries currently enrolled in the program. The system also provides a centralized database to ensure efficient record-keeping, reduce the manual burden of monitoring educational progress, and promote transparency throughout the scholar's lifecycle.

The scope of the project is limited to the management and monitoring of programs specifically administered by the Fr. Marie Eugene Family Development Foundation. It does not extend to scholarship initiatives managed by external NGOs, private institutions, or government agencies. While the system supports the recording of academic performance and session attendance, it does not include a dedicated native mobile application (iOS/Android), automated financial aid disbursement, or direct integration with third-party banking and payment gateway systems. Furthermore, the system requires a modern web browser and an active internet connection for operation, as offline functionality and synchronization are outside the current scope of development.

1.8 Definition of Terms

The following terms are defined conceptually and operationally as used within the context of this study:

Automated Flag / Digital Trigger. Conceptually, this term is a notification that is automatically generated by a computer program when a predetermined criterion is

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recognized. In MEFAMDEV-Life, it is a computer-generated alert that is automatically sent if a student's grade for a quarter is lower than a specified minimum threshold or if a scholar's Spiritual Formation attendance is less than 75% for a specified period, prompting Admin users to undertake action.

Beneficiary Family. Conceptually, this term is a household that receives welfare or development services from a social organization. In this study, it is a household among the families that are registered under MEFAMDEV's integral development programs, whose socio-economic data is being maintained by the MEFAMDEV-Life system.

Centralized Web Portal. A single interface that is accessible through the internet that integrates various data management activities into one interface. In this study, this term is used to refer to the MEFAMDEV-Life web application, which is the sole repository of all family/student data of the foundation that can be accessed through any internet-enabled device.

Integral Development. MEFAMDEV's philosophical approach to development that considers the simultaneous development of a person's physical, intellectual, economic, and spiritual aspects. This is the overarching framework that all programs of MEFAMDEV, including Education, Feeding, Livelihood, and Spiritual/Values Formation, are designed to operate under.

Recipient. The student-beneficiary who is currently in MEFAMDEV from elementary to Senior High School level, and is currently receiving a grant from the foundation. In the system, they are classified as external users, with limited access to view only their own academic status, attendance, and alert status.

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Scholar Lifecycle. The whole lifecycle of a student-beneficiary within the MEFAMDEV, from initial enrollment through the period of active monitoring, until the point of graduation from Senior High School. MEFAMDEV-Life is intended to document the whole lifecycle of the student.

Socio-Economic Profile. A structured profile of the family's living conditions, income levels, educational attainment of family members, and other indicators of their economic well-being. In the context of the study, it refers to the standardized family data kept in the MEFAMDEV-Life system, which serves as the base data for each family-beneficiary.

Spiritual Formation. A monthly program component of MEFAMDEV that consists of bible study sessions and values lessons that relate religious teachings to one's living conditions. In the system, scholars' attendance to Spiritual Formation is monitored as part of their eligibility criteria.

Web-Based Management System. An application that can be used by accessing the internet using a web browser without requiring local installation on individual devices. MEFAMDEV-Life is developed as a web-based system using React.js, Laravel, and MySQL to ensure the system can be accessed from any device by authorized users.

Year-over-Year Academic Growth Tracking. A longitudinal data monitoring approach used to monitor a scholar's academic performance records over the years. In MEFAMDEV, the Year-over-Year Academic Growth Tracking refers to the system's module that collects all the grade information for a scholar from the first year of

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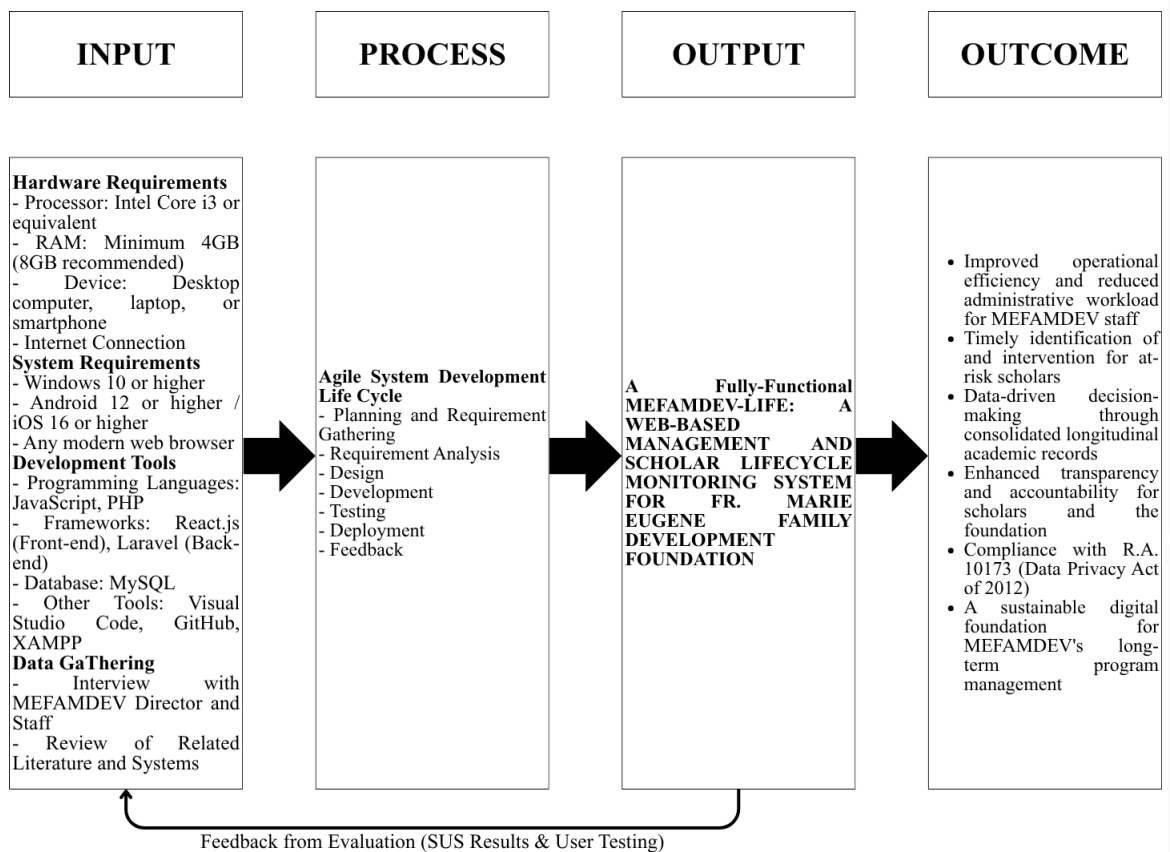
enrollment to the current year, enabling the detection of academic improvement or decline over time.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Conceptual Framework

Figure 2.1

IPO Conceptual Framework

2.2 Related Literature

Foreign Literature

The global landscape of social welfare and non-governmental organizations is undergoing a significant shift toward data-driven governance. Orero-Blat et al. (2024)

explored how advancements in information and communication technologies and data analytics are fundamentally transforming organizational decision-making. However, their research highlighted a persistent digital divide, noting that many sectors and regions still struggle to effectively utilize digital data for program management. They advocate for NGOs and foundations to adopt integrated information systems to move beyond fragmented, siloed records and toward more cohesive, data-driven strategies.

Building on the challenges of manual systems, Godefroid et al. (2023) reported that a substantial number of international foundations and NGOs continue to rely on paper-based processes for record-keeping and service delivery. Their findings suggest that these manual methods severely limit monitoring, learning, and accountability capacities. They argue that digital transformation—specifically the implementation of centralized data systems—is essential to reducing human error, accelerating reporting cycles, and enhancing coordination among frontline program staff.

The practical application of these systems is evident in multi-country studies on inclusive digital social protection. Qomariyah and Mursyidah (2025) further validated the effectiveness of digital social welfare systems at the grassroots level through their study of the Next Generation Social Welfare Information System (SIKS-NG) in Indonesia. Their evaluation found that the transition to a centralized e-government platform significantly improved the accuracy of social assistance targeting and data updating at the village level. However, the study also identified critical success factors that are highly relevant to the implementation of MEFAMDEV-Life, including the necessity of competent system operators, adequate digital infrastructure, and the challenge of

overcoming limited technical skills among staff. This research emphasizes that while the software provides the framework for efficiency, organizational commitment and continuous technical support are essential for transforming poverty alleviation strategies at the community level.

Furthermore, the technical feasibility of these platforms is further supported by research into web-based monitoring systems designed for academic integrity. Davis and Rahim (2025) demonstrated that integrating real-time monitoring features—such as automated activity detection and session time control—into a web-based exam system provides a viable and globally replicable solution for institutions requiring oversight without the complexity of AI-based tools. Their study found that a structured Waterfall development approach resulted in a highly reliable system where 100% of data was saved without loss and automated intervention (such as lockout mechanisms) was successfully triggered by specific user actions. This confirms that properly designed web applications can effectively support high-stakes monitoring processes while ensuring usability and system stability, even under standard internet bandwidth conditions.

Local Literature

In the Philippines, the necessity for structured beneficiary data has become a focal point for effective development work. Aguiling (2023) emphasized that accurate and accessible beneficiary information is the cornerstone of understanding participation levels and tailoring interventions within Philippine NGOs. The study demonstrates that the way an organization documents beneficiary profiles directly dictates how it classifies

stakeholders and designs participatory mechanisms, underscoring the critical role that a reliable information system plays in the success of social programs.

The technical viability and user acceptance of such digital transitions are further supported by recent local research. Gasmido et al. (2025) evaluated the implementation of "Webyu," a web-based management system designed to replace outdated manual processes in a local barangay. Using the Scrum framework for iterative development, the study proved that ICT-driven solutions could achieve exceptional performance ratings under the ISO 25010 software quality criteria—specifically in the areas of maintainability (4.76) and security (4.65). Furthermore, using the Technology Acceptance Model (TAM), the researchers found that perceived usefulness and ease of use were the primary drivers for adoption among residents and officials. These findings provide a scalable model for MEFAMDEV-Life, demonstrating that a well-designed web portal can successfully streamline administrative tasks, reduce processing times, and enhance data accessibility in a way that is both secure and user-friendly for non-technical staff.

Finally, Albano (2024) demonstrated that open-source tools—specifically PHP, HTML, JavaScript, and MySQL—are highly effective for building dynamic web-based systems that enhance transparency and public service delivery in the Philippines. The study's evaluation using a 5-point Likert scale showed that users highly valued a logical structure and easy navigation, proving that sophisticated, high-impact systems do not require expensive proprietary software. This directly validates the technical architecture of MEFAMDEV-Life, which utilizes a similar open-source stack to ensure the system

remains sustainable and maintainable for a community foundation without a large IT budget.

2.3 Related Studies and/or Systems

Foreign Studies and/or Systems

Sarjito (2025) evaluated the technical implementation of the Social Welfare Information System-Next Generation (SIKS-NG), a centralized e-government platform designed to modernize the management of social assistance programs in Indonesia. The study found that the system's ability to integrate fragmented records into a unified database significantly improved data centralization and transparency at the local level. However, the evaluation also highlighted that technical success is heavily dependent on infrastructure and the digital capacity of administrators—findings that directly inform the design of MEFAMDEV-Life's socio-economic profile module, which aims to move the foundation from fragmented manual records to a unified, data-driven architecture.

Amer (2025) developed and proposed a Smart Scholarship Management System (SMS) as an all-in-one web-based solution for educational organizations in Lebanon. The system features a robust technical architecture including role-based access control, budget tracking, and detailed student-donor profiling to automate administrative workflows. The study concludes that the implementation of such a platform fills significant gaps in service delivery by providing a transparent and accountable environment for monitoring underserved communities. The advanced tracking and reporting features of the Smart SMS directly parallel the technical objectives of

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MEFAMDEV-Life's scholar lifecycle monitoring and year-over-year academic growth tracking modules.

Local Studies and/or Systems

Blancaflor et al. (2022) developed and evaluated APPLY, an online tertiary-level scholarship management system designed to address the increasing volume of Need-Based Academic Scholarship (NBAS) applications in the Philippines. The system was designed as a web-based portal to eliminate time delays caused by manual processing and large volumes of physical requests. Using the Post-Study System Usability Questionnaire (PSSUQ), the researchers found that the system achieved high scores for usefulness and interface quality from both student applicants and scholarship administrators. The study's conclusion that web-based automation is vital for maintaining administrative efficiency mirrors the rationale for MEFAMDEV-Life, which seeks to replace manual audits with a functional, browser-accessible monitoring system.

Buena et al. (2026) introduced the Local Government Unit 2: Social Service and Welfare Management System, a digital platform built using Microservices Architecture to enhance social service delivery in the Philippine public sector. The system centralizes critical operations such as beneficiary management, case tracking, and real-time analytics to replace fragmented and inefficient manual workflows. Technically, the platform provides social workers with a dedicated "Case Portal" and automated alerts, which significantly reduces errors and promotes data-driven decision-making. The system's focus on providing real-time case access and a centralized beneficiary database serves as a direct local precedent for MEFAMDEV-Life's Socio-Economic Profile and Monitoring

modules, validating the use of centralized web technologies to foster transparency and responsiveness in grassroots organizations.

2.4 Synthesis

The reviewed local and foreign studies emphasize the importance of integrated, web-based systems in improving beneficiary and scholarship management through automation, centralized data handling, and real-time monitoring. These systems successfully address the inefficiencies of manual record-keeping and fragmented applicant tracking, providing a solid operational rationale for cloud-based platforms in both NGO and social protection settings. However, many existing systems focus only on individual components like basic application submission or isolated case tracking that lack a unified approach that combines comprehensive family profiling, academic growth tracking, and automated monitoring within a single platform.

To address these gaps, the proposed MEFAMDEV-Life introduces an integrated suite of modules designed for the specific operational workflows of a DSWD-registered social welfare foundation, including centralized scholar management and automated academic tracking. By utilizing an open-source stack of PHP, HTML, and MySQL, the system aims to enhance transparency, minimize administrative workloads, and provide a purpose-built, secure environment tailored to the unique mission and scale of MEFAMDEV.

CHAPTER 3

METHODOLOGY OF THE STUDY

3.1 Software Development Methodology

Figure 3.1

Agile Software Development Life Cycle Model

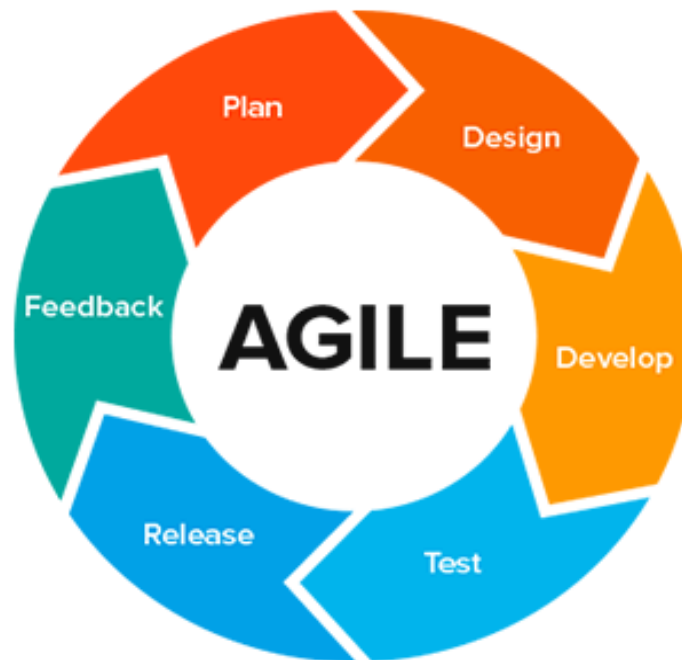


Figure 3.1 illustrates the Agile Model adopted in the development of the MEFAMDEV-LIFE: A WEB-BASED MANAGEMENT AND SCHOLAR LIFECYCLE MONITORING SYSTEM FOR FR. MARIE EUGENE FAMILY DEVELOPMENT FOUNDATION in Sta. Cruz, Angat, Bulacan. This approach presents a flexible, iterative approach that allows for continuous improvement, making it ideal for smoothly

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transitioning the Fr. Marie Eugene Family Development Foundation from manual paper and Excel-based records to a digital web portal.

During planning and gathering requirements, this phase involves identifying the operational challenges with the current manual setup by interviewing the Director and Staff. It helps in understanding and identifying its specific goals, such as tracking the academic progress of Basic Education beneficiaries, tracking the attendees at every event, monitoring the recipients of feeding programs, and digitizing the socio-economic profile of every beneficiary family. The developers then analyze the gathered data to define the exact technical and hardware needs, taking into consideration constraints like the foundation's older computer models and the need for a system accessible via modern web browsers.

In the Design phase, it focuses on developing user interface layouts and the database. Using MySQL as a database and HTML and CSS for designing the user interface layout, this system is designed to be user-friendly, responsive, and specifically tailored for the primary roles: Admin and Staff. The design allows Admin and Staff to upload and monitor their beneficiaries' requirements, such as:

1. Birth Certificate of the Students
2. Barangay Certificate of Indigency
3. Handwritten letter from the students
4. Proof of family household income
5. Certificate of Registration/Grade (COR/COG)

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The development phase involves the actual building and coding phase. This phase is implementing the system using JavaScript and React.js for the front-end, and PHP/Laravel for the back-end. The front-end provides the blueprint, design, and user interface, while the back-end is responsible for managing data storage, databases, authentication, and logic operations. The system's main modules development includes:

1. Family Management Module - Enables admins to view beneficiary family profiles to assess overall eligibility for the benefits the agency offers.
2. Scholar Management Module - Enables applicants to apply via online application and upload digital copies of required documents.
3. Grade Monitoring Module - Evaluates the applicant's grade per quarter for further improvements and interventions.
4. Attendance Monitoring Module - Evaluates the beneficiary's attendance for yearly reports.
5. Academic Growth Tracking Module - Enables admins to consolidate and track the academic progress of grade records across multiple grade levels and school years.
6. Report Generation Module - Enables Admins to generate reports such as academic performance and program outcomes for internal program assessments and external stakeholders reporting.
7. User Account Management Module - Enables administrators to view, add, edit, and delete existing /deactivated accounts for security purposes.
8. Admin Dashboard Module - Summary view of the total beneficiary families, active beneficiaries, and number of active alerts.

9. Self-Service Portal Module - Enables applicants to track the status of their application and view their academic credentials, quarterly grades, and attendance history.

During the Testing phase, the produced modules are verified to ensure that all functional requirements run smoothly and accurately. The accuracy and dependability of the system were tested in a number of ways. This feature involves testing the real-time correctness of the document and attendance monitoring, as well as confirming that the role-based access restrictions and automatic digital triggers for when a beneficiary requires intervention or has low grades successfully secure sensitive family data. In order to satisfy the system requirements, any errors and inconsistencies were noted and fixed right away.

After the Testing phase, the system will be deployed in a controlled environment for the initial evaluation by the Fr. Marie Eugene Family Development Foundation's (MEFAMDEV) staff and recipients to use. The fully functional web-based management and scholar lifecycle monitoring system handles the data of online applications, document uploads, real-time attendance tracking, and tracking of every spiritual formation accurately. The beneficiary and staff then tested the portal's reliability in submitting, verifying, and monitoring applications and information.

In the Feedback phase, once it is deployed, the developers then gather data from the beneficiary, admin, and staff based on their experience using the portal. Their insights emphasized the consistency of evaluating and tracking features, which allowed the admin and staff to monitor their programs and beneficiaries' status and to make an immediate

response needed. This phase allows developers to take necessary action to improve the system's performance based on the feedback of the users.

3.2 Requirement Specification

3.2.1 Functional Requirements

The functional requirements define the core features and operations that the proposed Web-Based Management and Scholar Lifecycle Monitoring System in Fr. Marie Eugene Family Development Foundation of Sta. Cruz Angat, Bulacan must ensure efficient scholar management, document evaluation, and tracking.

1. **User Registration and Authentication:** The system shall allow administrators and staff to register and log in securely using unique credentials. The system shall validate the user's credentials before giving access to the appropriate dashboard.
2. **Beneficiary Application Module:** The system shall allow new applicants to fill out and submit online application forms. The system shall enable users to upload the required documents. The system shall store and organize the submitted documents in a centralized database that only authorized staff can access.
3. **Tracking and Monitoring System:** The system shall provide admins with tools to track academic progress, application status, and attendance. Beneficiaries shall be able to view their application status through their online dashboard.
4. **Announcement Module:** The system shall allow admins to create and edit announcements.
5. **Administrator Management:** The system shall allow authorized staff to review, evaluate, approve, and update beneficiary applications. The system shall be able

to manage beneficiary records, generate summary reports, and update beneficiary data when needed.

3.2.2 Non-Functional Requirements

Non-functional requirements define the quality standards and constraints that the system must satisfy:

1. Performance: System pages must load within three seconds under normal internet conditions.
2. Security: All data transmissions must be secured via HTTPS/SSL encryption.
User passwords must be hashed. Role-based access control must be enforced for all system functions.
3. Usability: The system interface must be intuitive for non-technical social work professionals. Key tasks (e.g., encoding grades, viewing scholar profiles) must be completed within three user interactions from the dashboard.
4. Availability: The system must be accessible 24 hours a day, seven days a week, subject to scheduled maintenance windows communicated in advance to users.
5. Compatibility: The system must function correctly on current versions of Google Chrome, Mozilla Firefox, and Microsoft Edge on both desktop and mobile devices.
6. Scalability: The system must be capable of accommodating growth in beneficiary family and scholar records without requiring architectural changes.
7. Data Privacy Compliance: The system must comply with the provisions of the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), including data

minimization, access control, and secure storage of personally identifiable information.

3.3 Technical Feasibility

The proposed MEFAMDEV-Life: A Web-Based Management and Scholar Lifecycle Monitoring System is technically feasible as it utilizes reliable and widely used web technologies. The system requires a stable internet connection to ensure smooth and continuous access for authorized staff and recipients.

The system is developed using modern web technologies. The front-end is built using HTML, CSS, and React.js to create a responsive and user-friendly interface. The back-end uses Laravel (PHP framework) and MySQL database for efficient data processing, storage, and security. These technologies are widely supported, cost-effective, and suitable for developing scalable web-based systems.

The system is designed to manage beneficiary family records, recipient profiles, academic grades, and spiritual formation attendance in a centralized database. It also includes automated monitoring features that allow staff to identify recipients who are at risk based on their academic performance and attendance.

In terms of infrastructure, the system can operate on standard web hosting environments. A server with adequate processing power, memory, and storage is recommended to support multiple users accessing the system simultaneously. Overall, the proposed system is technically viable and can be implemented using available resources.

3.3.1 Compatibility Checking

Table 3.1*Software and Hardware*

Device / Platform	Software Requirement	Compatibility	Hardware Requirement	Compatibility
Minimum (Desktop / Laptop)	Windows 10 or higher	Yes	Intel i3, 4GB RAM, 128GB Storage	Yes
Recommended (Desktop / Laptop)	Windows 10 / 11	Yes	Intel i5, 8GB RAM, 256GB Storage	Yes
Minimum (Mobile)	Android 10 / iOS 13	Yes	3GB RAM, 32GB Storage	Yes
Recommended (Mobile)	Android 12+ / iOS 15+	Yes	4GB RAM, 64GB Storage or higher	Yes

3.3.2 Relevance of the Technology

The use of web-based technology in MEFAMDEV-Life is highly relevant in addressing the challenges faced by the foundation in managing beneficiary and recipient data. Traditional methods such as paper records and spreadsheets often result in delays,

data inconsistencies, and difficulty in tracking academic and attendance records over time.

By implementing a centralized web-based system, MEFAMDEV-Life improves data organization, accessibility, and security. Staff can efficiently manage records, monitor academic performance, and track spiritual formation attendance in real time. This reduces manual workload and minimizes errors in data handling.

The system also incorporates automated alerts that identify recipients with low grades or insufficient attendance. This allows staff to take timely action and provide necessary interventions to support the recipients. Additionally, the system supports report generation, enabling the foundation to evaluate program outcomes and make data-driven decisions.

Overall, the selected technologies are appropriate for the needs of MEFAMDEV as they are accessible, efficient, and capable of improving service delivery through a structured and reliable digital system.

3.4 Operational Feasibility

The proposed MEFAMDEV-LIFE: A Web-Based Management and Scholar Lifecycle Monitoring System is operationally feasible as it streamlines the recruitment, monitoring, and lifecycle management processes of the Fr. Marie Eugene Family Development Foundation. The system promotes efficiency, data integrity, and long-term sustainability by transitioning the foundation from manual, spreadsheet-based record-keeping to a centralized, structured digital platform.

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Staff members can conveniently manage the entire scholar lifecycle from initial application to graduation through a web-based interface that mirrors their existing data entry habits in Microsoft Excel, ensuring an incremental and manageable change in daily workflow. A key tracking feature enables administrators to monitor the real-time status of beneficiaries, providing immediate visibility into academic performance and scholarship standing without the need for manual cross-referencing of multiple files.

For the MEFAMDEV office in Angat, Bulacan, the system provides an organized way to review scholar requirements and track historical data directly within a secure environment. While the final verification of documents remains a manual human process, the centralized access to digitized records significantly simplifies administrative oversight. The integration of automated database backups and scheduled framework updates ensures the system remains reliable and secure with minimal technical intervention from the foundation's staff.

The system's mobile-responsive design further enhances operational flexibility, allowing staff to access and update scholar information during field-based monitoring via any internet-connected device. This accessibility reduces dependence on office-bound hardware and ensures that the foundation can maintain accurate, up-to-date records regardless of their physical location, ultimately improving the responsiveness and transparency of the scholarship program.

3.5 Schedule Feasibility

3.5.1 Gantt Chart

Table 3.2

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Gantt Chart

PROCESS	NOVEMBER 2025				DECEMBER 2025				JANUARY 2026				FEBRUARY 2026				MARCH 2026			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Planning																				
Requirement Gathering																				
Documentation																				
Design																				
Development																				
Test																				
Deployment																				

3.6 Economic Feasibility

3.6.1 Cost and Benefit Analysis

Table 3.3

One-Time Development Cost

Item	Cost	Justification
Labor / Development	₱12,000 (proposed)	Developed by student researchers as a capstone project; no external developer fees
Software & Dev Tools	₱0	React.js, Laravel, PHP, MySQL, VS Code, and GitHub are all free and open-source
System Documentation & Printing	₱500 - ₱1,500	Printing of user manuals, training materials, and research output

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Training & Orientation (Staff)	₱0 - ₱500	Conducted by the research team during deployment; no external trainer needed
TOTAL	₱12,500 - ₱14,000	

Table 3.4

Recurring Annual Operational Cost

Item	Cost	Justification
Cloud Hosting / Web Server	₱0 - ₱7,000	Free on academic tiers (e.g., Railway, Render); minimal fee on basic shared hosting
Domain Name Registration	₱500 - ₱1,500	Optional custom domain (e.g., mefamdev.org); not required if using a free subdomain
Internet Access	₱0	Already part of the foundation's existing office setup
Hardware (Computers, Phones)	₱0	Existing donated devices are sufficient per compatibility requirements
System Maintenance &	₱0	Handled by the foundation's designated admin using the open-source stack

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Updates

TOTAL ₱500 - ₱8,500

Table 3.5

Total Estimated Cost Summary

Phase	Cost
One-Time Development Cost	₱12,500 - ₱14,000
Recurring Annual Operational Cost	₱500 - ₱8,500 (Annually)
TOTAL	₱13,000 - ₱22,500

3.7 Requirements Modeling

3.7.1 System Flowchart

Figure 3.6

Admin/Staff Flowchart

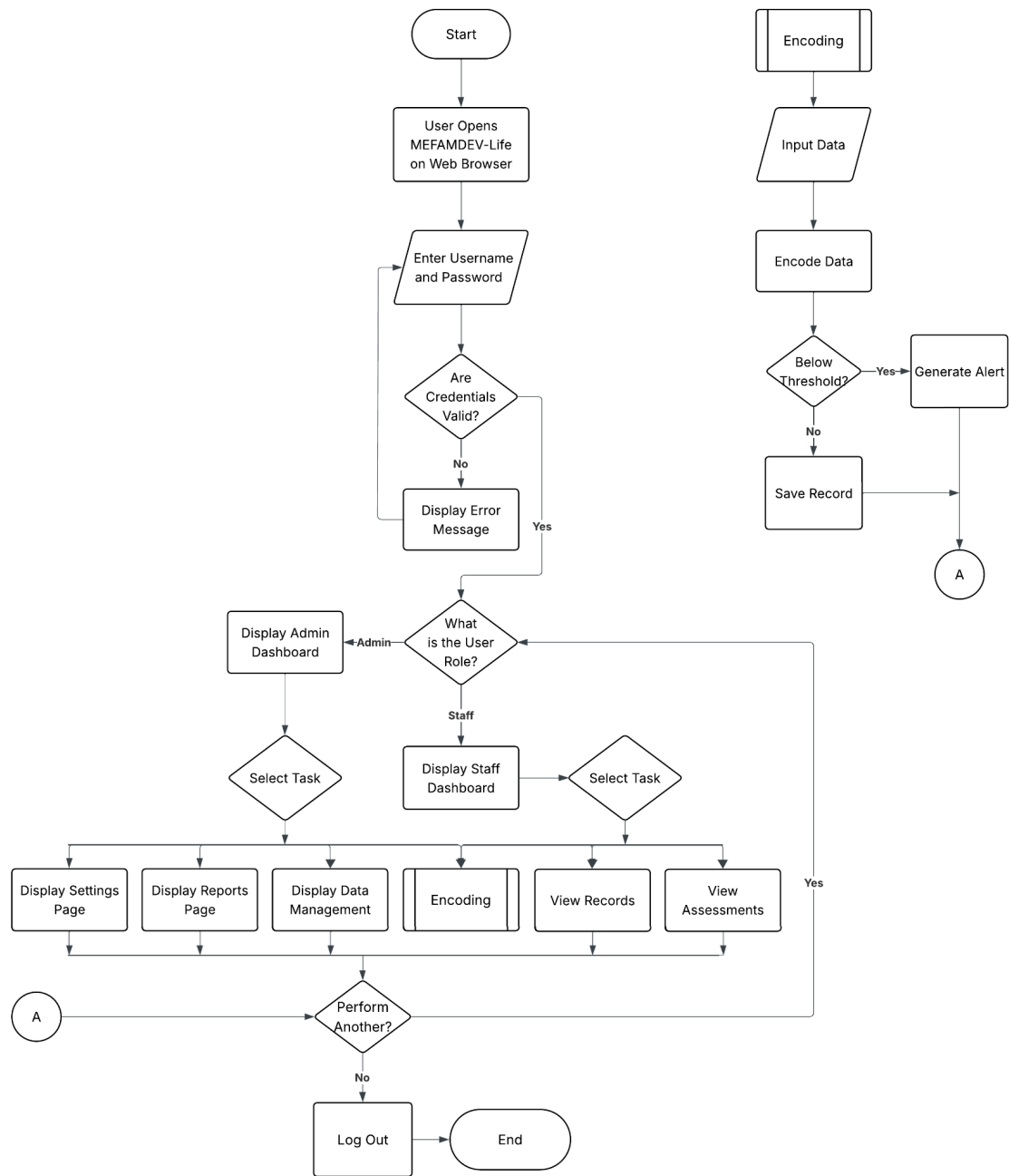
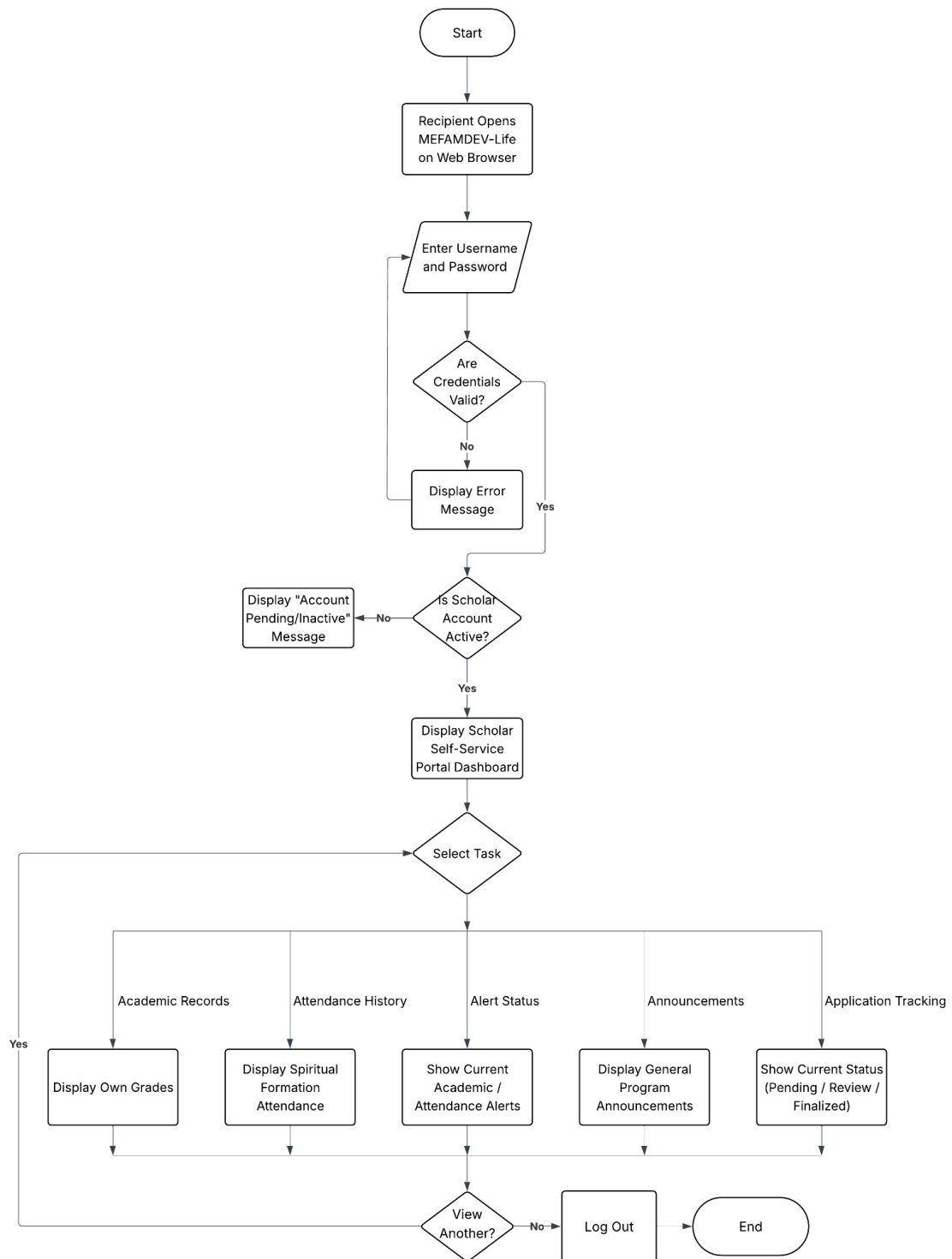


Figure 3.7

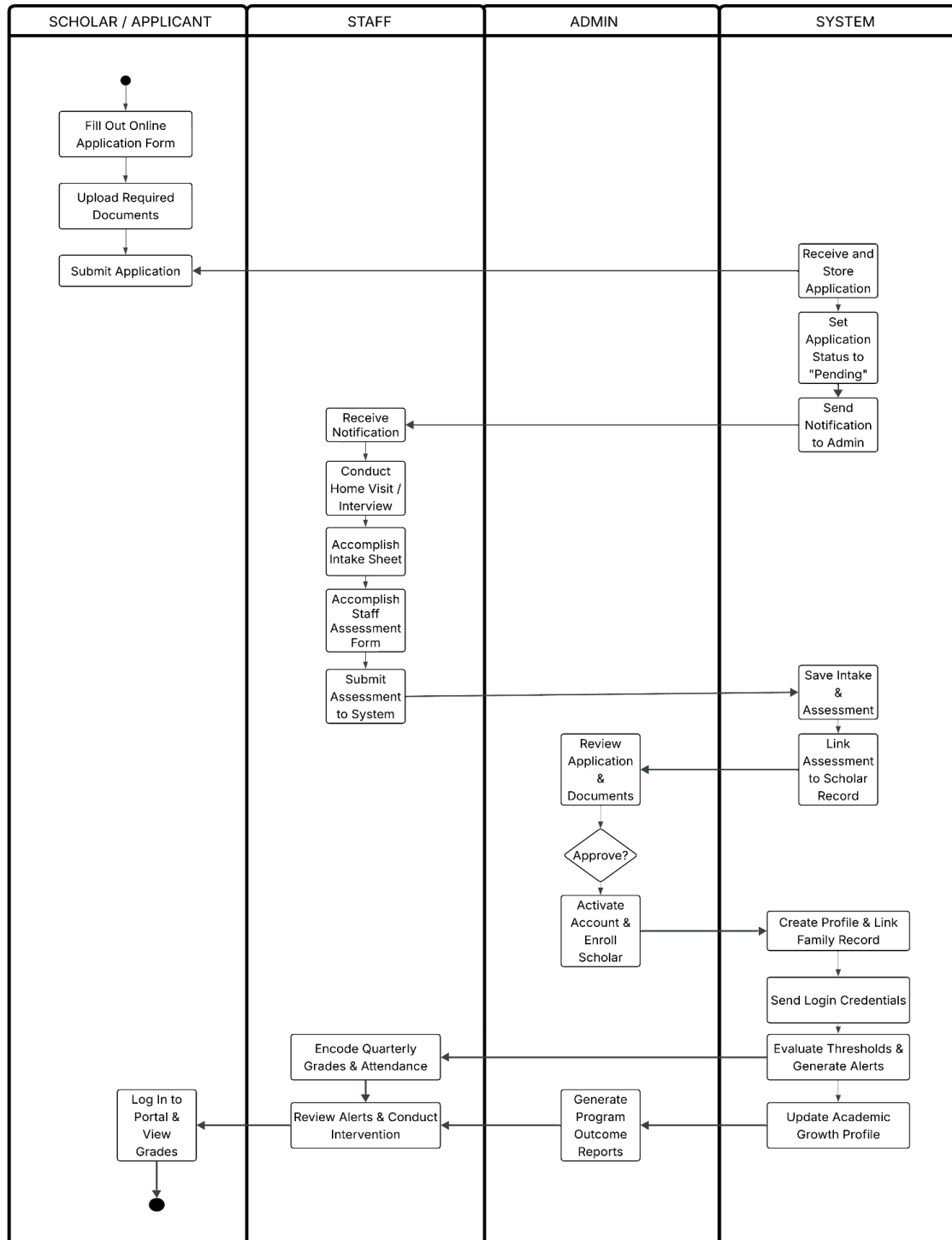
Recipient Flowchart

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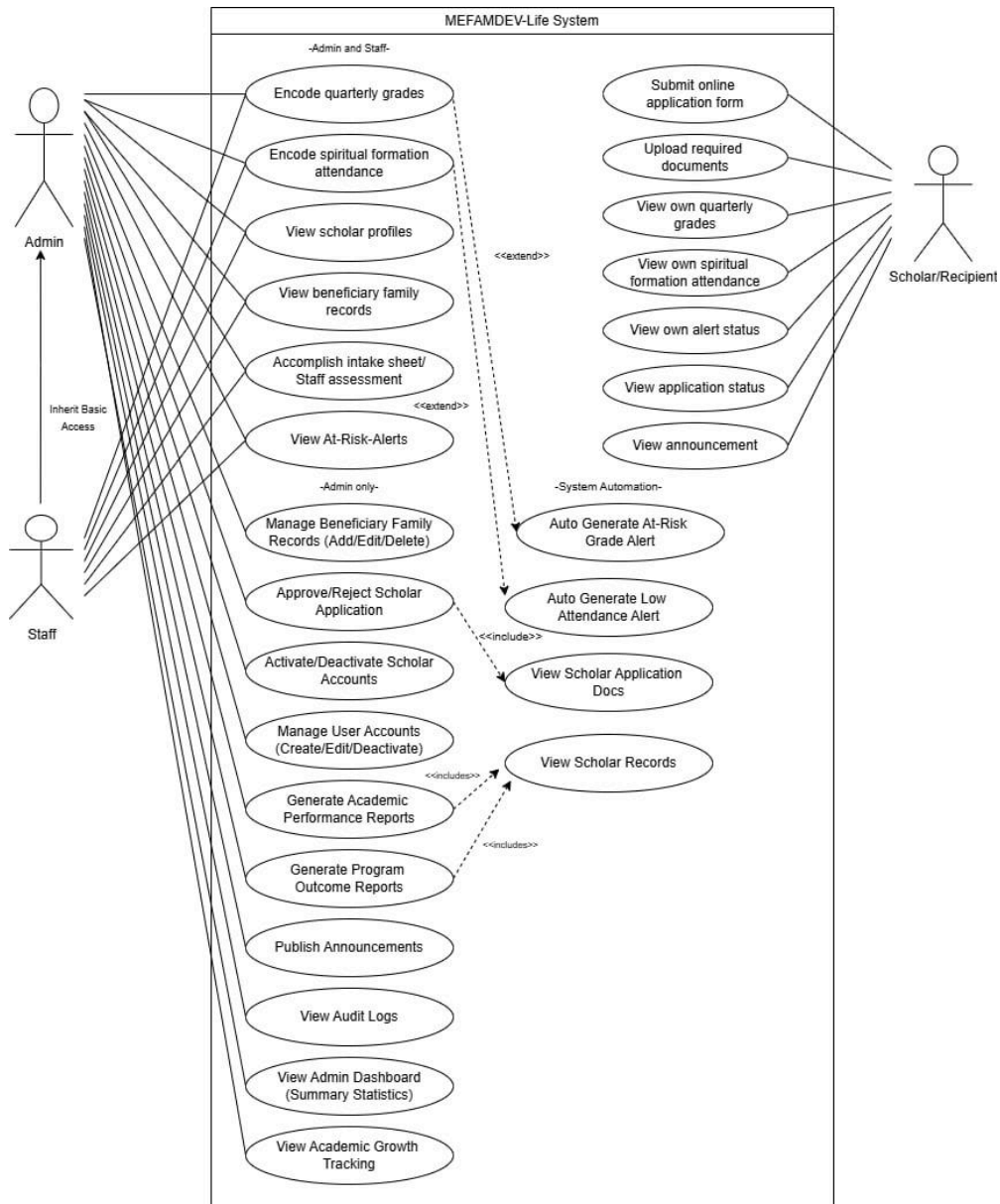
3.7.2 Activity Diagram

Figure 3.8



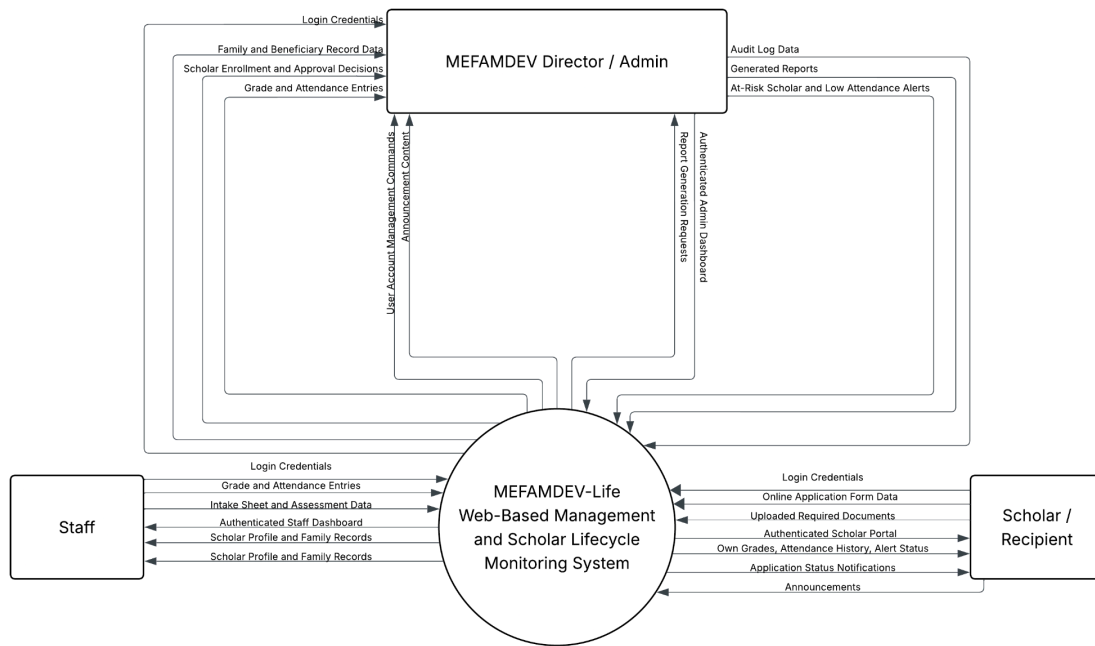
3.7.3 Use Case Diagram

Figure 3.9



3.7.4 Context Diagram (Level 0 DFD)

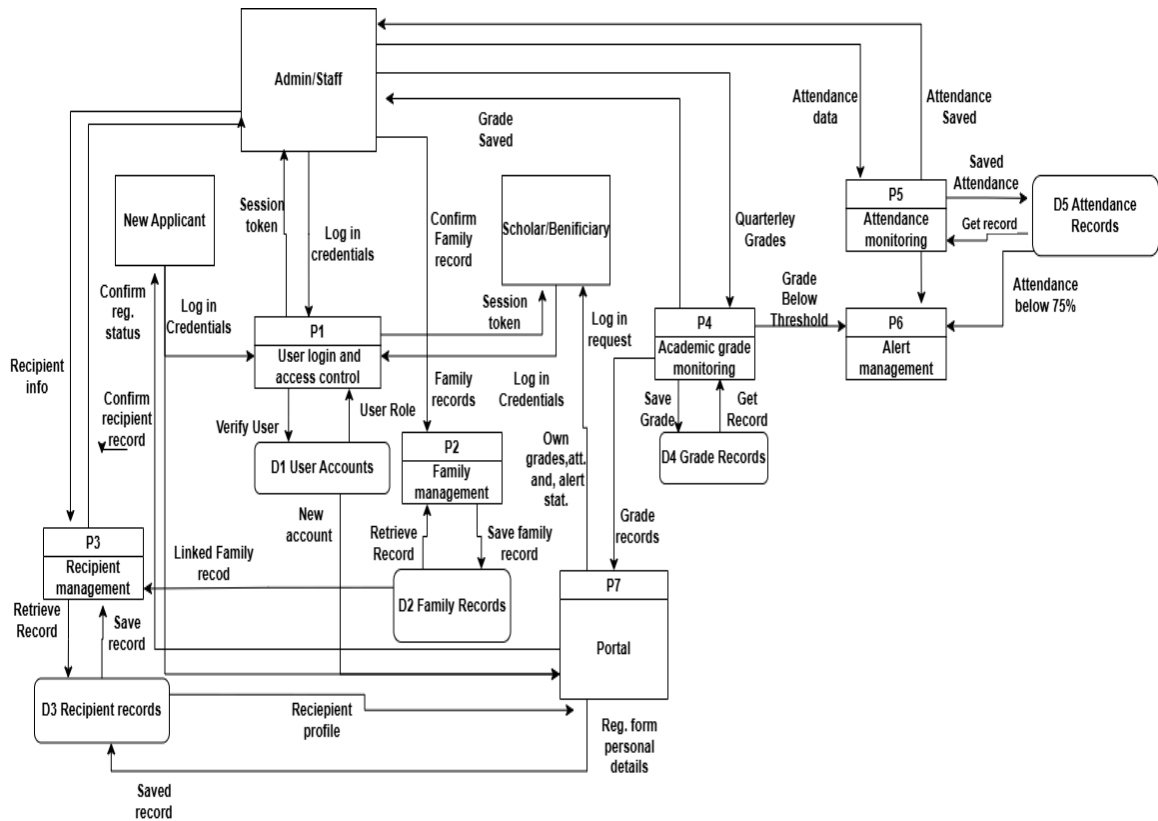
Figure 3.10



3.7.5 Data Flow Diagram (Level 1)

Figure 3.11

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3.8 Risk Analysis, Risk Assessment, and Risk Management

Table 3.6

Risk	Probability	Impact	Control Measures
Server downtime	Low	High	Use a reliable hosting provider and backups
Data loss	Low	High	Perform regular database backups

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Staff resistance	Medium	Medium	Provide training and a user-friendly interface
Poor internet connection	High	Medium	Allow simple and quick encoding during field work
Team delays	Medium	Medium	Proper task distribution and timeline management

3.9 User Interface Design

Figure 3.12

Login Page

The screenshot shows a web browser window with the address bar displaying 'login.html'. The page is divided into two main sections. On the left, there is a dark blue sidebar with a large yellow circle containing the letter 'M' at the top. Below this, the text 'MEFAMDEV-Life System' is displayed, followed by 'Fr. Marie Eugene Family Development Foundation' and 'Angat, Bulacan Chapter'. There are three menu items: 'Dashboard with real-time analytics', 'Scholar profiles and grade tracking', and 'Applicant portal and status tracking'. On the right, the main content area has a 'WELCOME BACK' message, followed by 'Sign in to your portal' and 'Choose your access type below'. There are two buttons: 'Staff / Admin' and 'Applicant'. Below these are input fields for 'USERNAME' and 'PASSWORD'. A 'Sign In to Admin Dashboard -->' button is present. At the bottom, there is a 'Demo Accounts:' section with a list of accounts: 'director / director123 - Full Access', 'edu / edu123 - Education Staff', 'finance / finance123 - Finance Officer', and 'program / program123 - Program Coordinator'.

Figure 3.9

Public Application Form

The screenshot shows a web browser window with the address bar displaying 'application_form.html'. The page has a dark blue header with a yellow circle containing the letter 'M' and the text 'EDUCATIONAL ASSISTANCE PROGRAM', 'Fr. Marie Eugene Family Development Foundation, Inc.', and 'Application Form para sa SY [2026-2027 *]'. Below the header, there are tabs for 'Tagalog' and 'English'. A progress bar shows six steps: 'Personal', 'Pamilya', 'Pinaayal', 'Edukasyon', 'Pangako', and 'Isuporta'. The 'Personal' tab is active. The form contains several input fields: 'BUONG PANGALAN' (with a placeholder 'Apelyido, Pangalan, GI'), 'HOUSE NO. / KALYE / PUROK', 'BARANGAY' (with a dropdown menu 'Pumili ng Barangay...'), 'ZIP CODE' (with a placeholder '3012'), 'KAARAWAN' and 'EDAD' (with a dropdown menu 'Pumili ng Kaarawan...'), and 'KASARIAN' (with a dropdown menu 'Pumili ng Kasarian...'). A 'Susunod -->' button is at the bottom right.

Figure 3.13

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Admin Dashboard

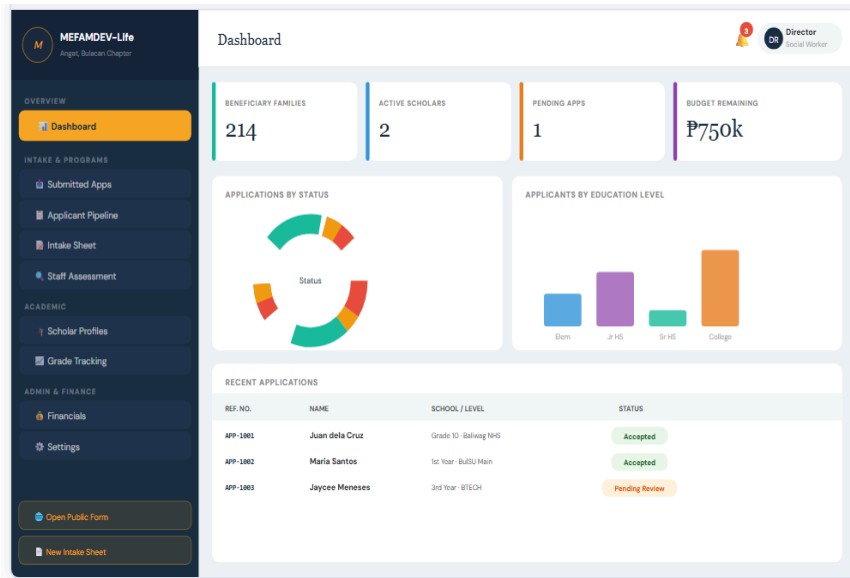


Figure 3.14

Applicant Portal

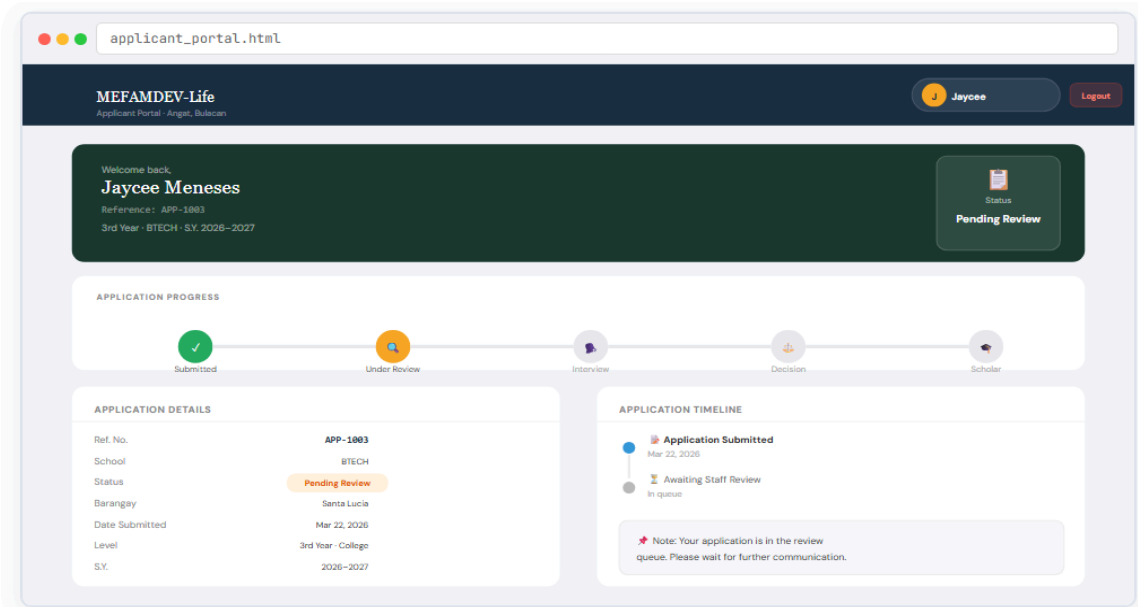


Figure 3.15

Intake Sheet & Staff Assessment (Staff-Only)

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FR. MARIE EUGENE FAMILY DEVELOPMENT FOUNDATION

Intake Sheet

Social Case Study Record

Auto-fill from application:
Select Applicant: *

CASE NO.
DATE
CASE CATEGORY
REFERRAL

I. IDENTIFYING INFORMATION

NAME
NICKNAME
PRESENT ADDRESS

II. FAMILY COMPOSITION

Name	Age	Sex	Relation	Educ	Occupation	Income

III. PROBLEM PRESENTED

Describe the presenting problem or need of the client...

FR. MARIE EUGENE FAMILY DEVELOPMENT FOUNDATION

Assessment of Staff

✓ = Positive / True X = Negative / False

KITA (Income)

1	May regular na trabaho o laging may pumapasok na kita	✓	×
2	Sapat ang kita para sa pamilya	✓	×

EDUKASYON (Education)

3	Nakapagtapos ang lahat ng mga anak sa elementarya	✓	×
---	---	---	---

KALUSUGAN (Health)

6	Nakakainin ng 3-5 beses sa isang araw ang pamilya	✓	×
---	---	---	---

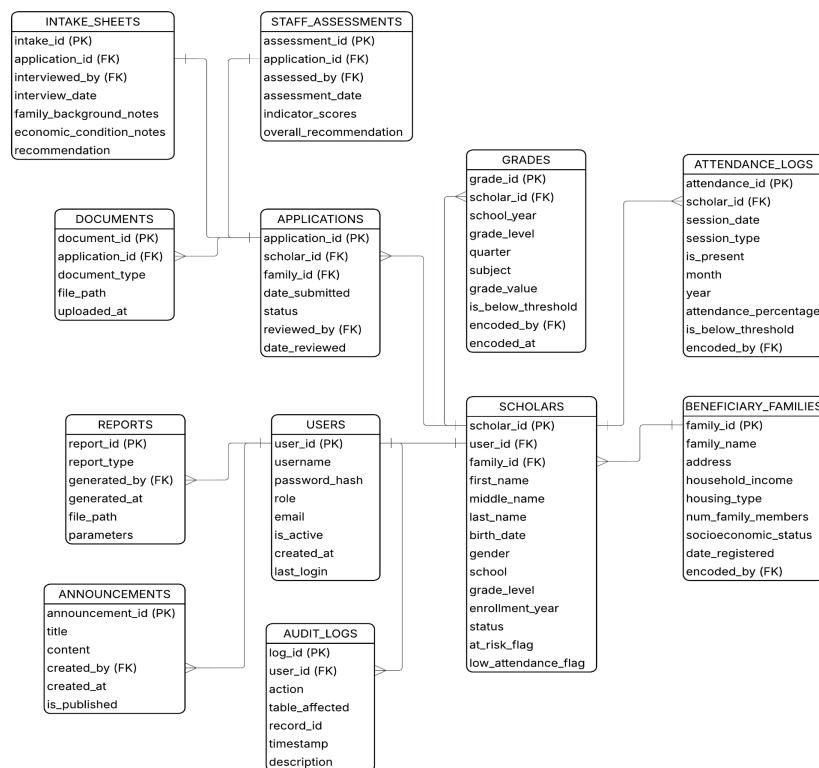
Assessment Results

9 Checks (-)
2 Crosses (X)
1 Starred X

BELOW POVERTY THRESHOLD
1 starred Indicator X

3.10 Entity Relationship Diagram

Figure 3.16



3.11 System Architecture

3.11.1 Network Model

Figure 3.17

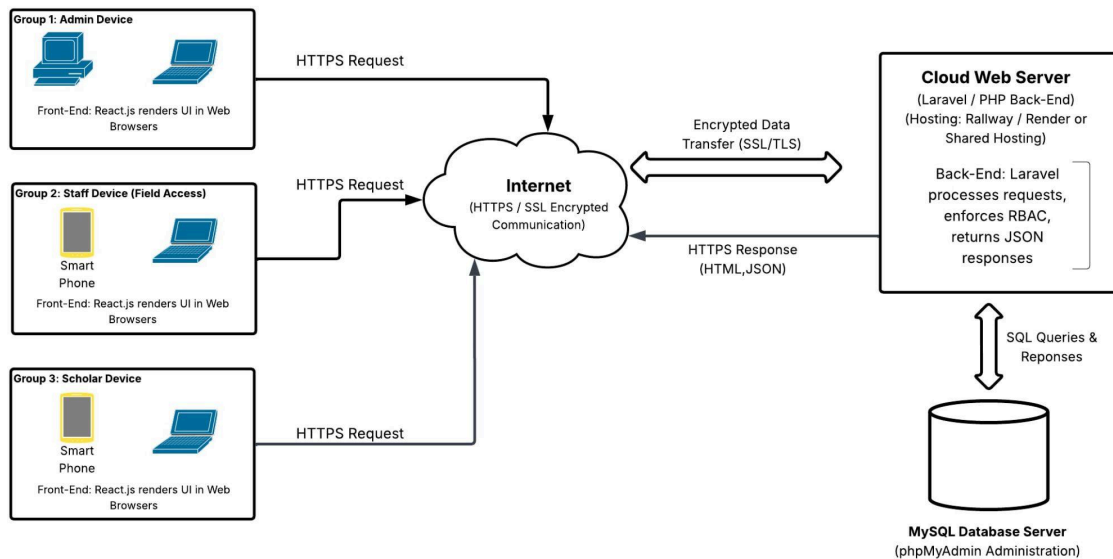


Figure 3.17 illustrates MEFAMDEV-Life and how data flows between user devices, the internet, the cloud server, and the database. Admin, Staff, and Scholar users may access the system through either a laptop or a smartphone. The Admin and Staff users interact with the administrative functions of the system, managing scholar records, family profiles, grades, and attendance, while scholars interact with the scholar portal provided to them to view their own academic information. The cloud server serves as the central hub that processes all user requests through the internet.

3.11.2 Network Topology

Figure 3.18

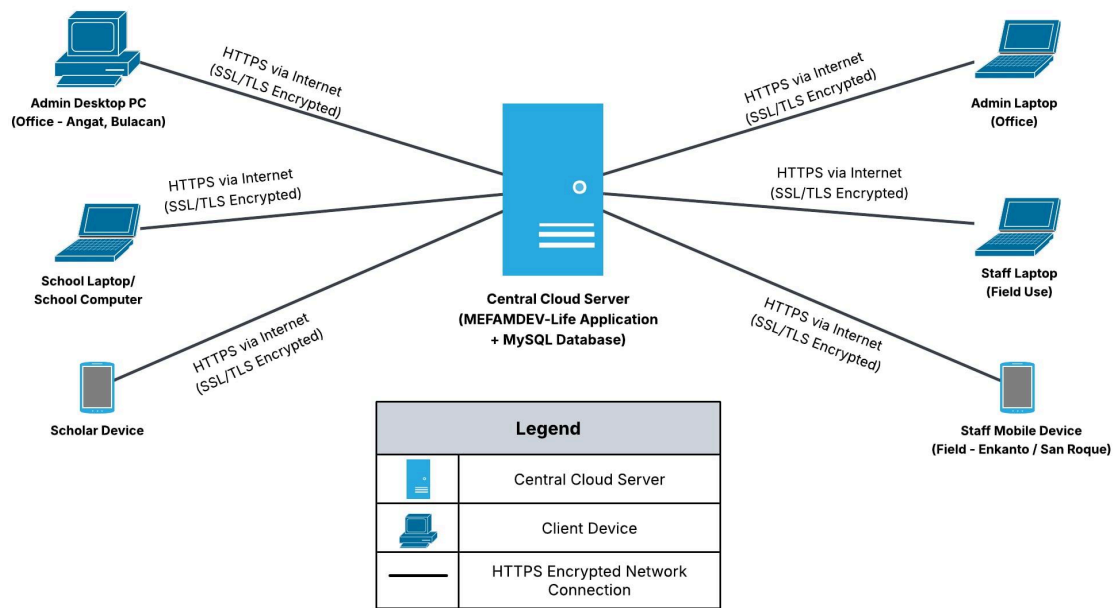


Figure 3.17 indicates MEFAMDEV-Life follows a Client-Server Star Network Topology in which all client devices, including Admin desktops, Staff laptops and mobile phones used during field visits, and Scholar devices, connect independently to a single centralized cloud server through the internet using HTTPS with SSL/TLS encryption, ensuring that all data transmitted between clients and the server is protected at all times.

3.11.3 Information Security

MEFAMDEV-Life was developed following standard security practices to ensure the protection of scholar, family, and foundation staff data. The system employs HTTPS encryption for all data transmissions, encrypts password hashing for stored credentials, and role-based access control to prevent unauthorized access — allowing Admin and Staff users to manage all records while scholars are restricted to viewing only their own information. Login attempt limitations and automatic session timeouts are implemented

to further safeguard against unauthorized use of unattended devices. Audit logging is also in place to record all data modifications, maintaining the integrity and traceability of all records within the system. These measures collectively ensure that all beneficiary and operational data is handled securely and in compliance with Republic Act No. 10173, also known as the Data Privacy Act of 2012, reducing the risk of unauthorized access, data manipulation, and privacy breaches.

3.12 Development and Testing

3.12.1 Software Specification

This section specifies the software and scripting languages used during the development of MEFAMDEV-Life, a web-based scholar lifecycle monitoring and family management system.

Table 3.7

Development Tools

Tool	Purpose
Visual Studio Code	Primary code editor IDE.
CSS	Styling and visual presentation of layout, formatting, and design.
React.js	Front-end JavaScript library for system's user interface.
Github	Remote hosting and management of the repository.

phpMyAdmin	Visual database administration.
Laravel	Primary back-end PHP framework
HTML	Front-end structure and content markup.
MySQL	Storing, managing, and retrieving all structured data.

3.12.2 Hardware Specification

This section defines the minimum and recommended hardware requirements for the system to function efficiently.

Table 3.8

Devices

Component	Minimum Requirement
Processor (laptop/desktop)	Intel Core i3 or AMD equivalent · 1.6 GHz or higher
Operating System	Windows 10 or Higher
Memory (RAM)	4GB Ram (Desktop/Laptop) / 2GB RAM (mobile devices)
Storage	No local storage required
Display	1280 × 720 pixels (recommended)

Internet Connection	1 Mbps stable connection
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Device Type	Desktop, laptop, and mobile browser (android 10 or higher/ ios 16 or higher)
-------------	---

3.12.3 Programming Environment

CSS. Responsible for the styling and visual presentation of MEFAMDEV-Life, ensuring consistent layout, formatting, and design across all pages of the system.

GitHub. Functions as the remote repository platform for the project's source code to be stored, backed up, and managed, ensuring the codebase was secured and accessible throughout the development process.

HTML. Establishes the structural foundation of the web pages, defining the layout, sections, and content elements of the MEFAMDEV-Life interface.

Laravel. Serves as the primary back-end PHP framework, providing the MVC architecture, routing, REST API handling, and middleware used for all server-side operations, authentication, and session management of the system.

phpMyAdmin. Provides a graphical user interface for managing the MySQL database, simplifying database administration, query execution, table inspection, and data handling during development.

React.js. Serves as the front-end JavaScript library for building the system's user interface through a component-based architecture, enabling fast rendering and reusable UI elements across all pages.

Visual Studio Code. Serves as the primary Integrated Development Environment (IDE), providing code editing, debugging, and project management functions for both front-end and back-end development, supported by extensions for PHP, and React.js.

MySQL. Acts as the Relational Database Management System (RDBMS) responsible for storing, managing, and retrieving all structured data of the system, including scholar records, family profiles, grades, attendance logs, and user accounts.

3.12.4 Test Plan

The Test Plan for MEFAMDEV-Life outlines the testing approach used to verify that the system functions correctly, meets its specified requirements, and is ready for use by the foundation's staff and scholars. The goal is to ensure that all core features are reliable, secure, and usable before the system is formally deployed.

Table 3.9

Test Case

Function	Test Case	Result Expected
User Authentication	Login with valid credentials	Redirect to correct dashboard based on role
User Authentication	Login with invalid credentials	Error message shown · login denied
User Authentication	Scholar accesses admin system	Access

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		denied/blocked
Scholar Registration	Submit complete registration form	Status set to Pending admin notified
Scholar Registration	Admin approves registration	Scholar record created
Scholar Monitoring	Enter failing grade	Scholar flagged as At-Risk
Spiritual formation attendance monitoring	Attendance drops below 75%	Low attendance alert raised on recipient profile
Family Management	Add new family record	Record saved and registered
Family management	Delete record	Record permanently removed from database
Account management	Admin creates new user account	Account created user can log in
Account management	Admin deactivates user account	Account disabled user cannot log in

3.13 Implementation and Evaluation

The implementation of MEFAMDEV-Life will begin with the deployment and configuration of the system on the foundation's cloud hosting server and its accessibility through standard web browsers on both desktop and mobile devices. Three types of accounts will be created: Admin, Staff, and Scholar.

Administrators and staff are responsible for managing family records, enrolling and monitoring scholars, encoding grades and attendance, and maintaining user accounts within the system. The automated at-risk flagging feature assists the staff in identifying scholars whose academic performance or attendance falls below the required threshold, reducing the burden of manual monitoring and minimizing the risk of oversight during evaluation.

Scholars can register through the scholar portal by submitting their personal, family, and academic information for admin review and approval. Once their account is activated, scholars can log in to view their own profile, grades, attendance records, and system notifications, providing them with transparent access to their academic standing within the foundation's program.

Before full deployment, the foundation's staff will undergo training and orientation to ensure proper and confident use of the system. A user manual will also be provided as a reference guide for navigating the system's features and performing common tasks during day-to-day operations and off-site community visits.

The evaluation phase will ensure that MEFAMDEV-Life meets the functional and usability requirements of the Marie Eugene Family Development Foundation, providing a

reliable, secure, and accessible platform for both staff and scholars. Evaluation will be conducted using the System Usability Scale (SUS) administered to the foundation's staff and selected scholars as respondents, measuring whether the system achieves the usability standards necessary for practical use within the foundation's operations. Continuous feedback gathered during and after evaluation will be used to identify areas for improvement and ensure the long-term accuracy, efficiency, and reliability of the system.

3.14 Ethical Considerations

The development of MEFAMDEV-Life involved careful consideration of the ethical responsibilities associated with handling personal and sensitive data belonging to the foundation's scholars, families, and staff. The following measures were observed throughout the research and development process to ensure the protection of all involved parties and to uphold the ethical standards required in academic research.

Compliance with R.A. 10173 — Data Privacy Act of 2012. The system was designed in compliance with Republic Act No. 10173, MEFAMDEV-Life implements several technical and organizational measures to uphold data privacy, including HTTPS encryption for all data transmissions, bcrypt password hashing for user credentials, role-based access control that restricts data visibility based on user roles, session timeout policies to prevent unauthorized access from unattended devices, and audit logging to maintain a traceable record of all data modifications within the system.

Confidentiality of Beneficiary Data in Research Output. In accordance with ethical research standards and the provisions of the Data Privacy Act, no personally

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identifiable information belonging to the foundation's scholars or their families will appear in any published or publicly accessible research output.

Informed Consent and Permission from Institution. Prior to the commencement of data gathering and system development, a Letter of Intent (LOI) was formally prepared and submitted to the Marie Eugene Family Development Foundation. The letter was signed by the research adviser and acknowledged by the foundation, granting the research team official permission to conduct the study, gather relevant information about the foundation's operations, and develop the proposed system on behalf of the organization

Transparency. The foundation's director and staff were informed of the purpose, scope, and limitations of MEFAMDEV-Life prior to its implementation. The nature of the data collected, how it will be used, who has access to it, and the security measures in place to protect it were made clear to all involved parties.

Integrity. Strict data validation measures applied on all form inputs, an audit log records all data modifications along with timestamps and the identity of the user who made the changes, providing a reliable trail that supports the detection and correction of any unintended alterations to scholar, family, or user records.

Accountability. The research team acknowledges the responsibility in ensuring that the system is developed with integrity, accuracy, and genuine intent to address the foundation's operational needs. The development and implementation of MEFAMDEV-Life is carried out with full accountability to the Marie Eugene Family Development Foundation and its beneficiaries. The foundation's designated administrator

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	is accountable for the proper use and management of the system after deployment, including overseeing user access, maintaining data accuracy, and ensuring that the system is used only for its intended purpose.	
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